Applying Interferometer-Based Dense Unitary Synthesis to Block-Diagonal Controlled Unitaries

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Abstract

This self-contained note gives a constructive, equation-level recipe for applying an interferometer-based dense unitary synthesis method to a block-diagonal controlled unitary

$$W = \sum_{x=0}^{M-1} |x\rangle\langle x| \otimes U_x, \qquad U_x \in \mathbb{C}^{N\times N}.$$

It first explains the beamsplitter/interferometer picture from scratch, then shows how to use the same rectangular interferometer skeleton for every block U_x, while making the QROM-loaded phase angles depend on the selector x. The selector register is never mixed with the target register; it is used only as an address for phase data. This exploits the block structure and avoids synthesizing W as a generic $MN \times MN$ dense unitary.

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1 Beamsplitters and interferometers: the basic idea

This section introduces the optical/interferometer language used in the synthesis. No physics of photons is required. Mathematically, a "mode" is just a basis state of the target register. A beamsplitter is a two-dimensional unitary that mixes two basis states while leaving all other basis states unchanged.

1.1 Modes as computational basis states

Let the target register have dimension $N = 2^n$ with computational basis

$$|0\rangle, |1\rangle, \ldots, |N-1\rangle.$$

In the interferometer language, each basis state $|y\rangle$ is called a mode. A general state is

$$|\psi\rangle = \sum_{y=0}^{N-1} c_y |y\rangle. \tag{1}$$

A unitary $U \in \mathbb{C}^{N\times N}$ transforms the vector of amplitudes

$$(c_0, c_1, \ldots, c_{N-1})^T \quad \longmapsto \quad U(c_0, c_1, \ldots, c_{N-1})^T. \tag{2}$$

An interferometer decomposition is a way to build this large unitary by repeatedly mixing only two modes at a time.

1.2 A two-mode beamsplitter

A beamsplitter acting on modes p and q is an $N \times N$ unitary that is identity on all modes except the two-dimensional subspace

$$\mathrm{span}\{|p\rangle, |q\rangle\}.$$

On that subspace it applies some 2×2 unitary G. In block form,

$$B_{p,q}(G) = \mathbb{I}_{\{0,\ldots,N-1\}\setminus\{p,q\}} \oplus G_{(p,q)}. \tag{3}$$

Equivalently, if

$$G = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \tag{4}$$

then

$$B_{p,q}(G)|p\rangle = a|p\rangle + c|q\rangle, \tag{5}$$
$$B_{p,q}(G)|q\rangle = b|p\rangle + d|q\rangle, \tag{6}$$
$$B_{p,q}(G)|r\rangle = |r\rangle \qquad (r \notin \{p,q\}). \tag{7}$$

So a beamsplitter is just a two-level unitary on the target basis.

1.3 A concrete rotation beamsplitter

A simple real beamsplitter is a rotation by angle θ:

$$R_y^{(2)}(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}. \tag{8}$$

Acting on two amplitudes, it gives

$$\begin{bmatrix} c_p' \\ c_q' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} c_p \\ c_q \end{bmatrix}. \tag{9}$$

A complex beamsplitter also includes phases. A common parameterization of a generic 2×2 unitary, up to an overall phase, is

$$G(\theta, \phi, \lambda) = \begin{bmatrix} \cos\theta & -e^{i\lambda}\sin\theta \\ e^{i\phi}\sin\theta & e^{i(\phi+\lambda)}\cos\theta \end{bmatrix}. \tag{10}$$

The synthesis method does not rely on this exact convention; it only needs the fact that every two-mode unitary can be expressed using phase shifts and fixed 50/50 mixing operations.

1.4 The 50/50 beamsplitter and the Hadamard

The special 50/50 beamsplitter mixes two modes equally. Up to phase conventions, it is the Hadamard matrix

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}. \tag{11}$$

In a quantum circuit, a Hadamard on the active two-mode qubit is a Clifford gate. Therefore, in a Toffoli-count cost model, it is effectively free. This is the source of much of the smaller constant in the interferometer-based method: turn arbitrary two-mode mixing into many free Hadamards plus diagonal phases.

1.5 Phase shifters

A phase shifter on one mode has the form

$$P(\alpha) = \begin{bmatrix} e^{i\alpha} & 0 \\ 0 & 1 \end{bmatrix}, \tag{12}$$

or, with a symmetric Z-rotation convention,

$$R_z(\alpha) = \begin{bmatrix} e^{-i\alpha/2} & 0 \\ 0 & e^{i\alpha/2} \end{bmatrix}. \tag{13}$$

These conventions differ only by global and one-sided phase bookkeeping. In the notes below we mostly use

$$R(\alpha) = \begin{bmatrix} e^{i\alpha} & 0 \\ 0 & 1 \end{bmatrix}. \tag{14}$$

1.6 Decomposing a two-mode unitary into phases and Hadamards

Any 2×2 unitary can be written as diagonal phases surrounding a small number of fixed 50/50 mixers. A useful schematic form is

$$G = D_0\, H\, D_1\, H\, D_2, \tag{15}$$

where

$$D_k = \begin{bmatrix} e^{ia_k} & 0 \\ 0 & e^{ib_k} \end{bmatrix}. \tag{16}$$

Because the D_k are diagonal, they can be implemented by phase rotations. Because the H gates are Clifford, the non-Clifford cost is concentrated in the phase-loading and phase-rotation machinery.

One way to see why Eq. (15) is plausible is parameter counting. A 2×2 unitary has four real degrees of freedom: one global phase and three $SU(2)$ parameters. The diagonal phases in D_0, D_1, D_2 provide enough phase degrees of freedom, while the two fixed Hadamards provide the required basis changes that convert diagonal phases into amplitude mixing.

1.7 From one beamsplitter to a layer of disjoint beamsplitters

If beamsplitters act on disjoint pairs, they commute and can be applied in parallel. For example, an even layer is

$$B^{\text{even}} = G_0 \oplus G_1 \oplus \cdots \oplus G_{N/2-1}, \tag{17}$$

where G_j acts on modes $(2j, 2j+1)$. In qubit notation,

$$y = 2j + s, \qquad s \in \{0, 1\}, \tag{18}$$

so an even layer applies a one-qubit operation to the least significant target bit s, with parameters controlled by the remaining bits j:

$$|j\rangle\, |s\rangle \longmapsto |j\rangle\; G_j\, |s\rangle\,. \tag{19}$$

This is exactly the structure that QROM likes: use j as an address, load the angles for G_j, apply the one-qubit phase/Hadamard sequence, then uncompute the loaded angles.

1.8 Alternating even and odd layers

Even layers only couple $(0, 1), (2, 3), \ldots$. To build a general dense unitary, we also need to couple neighboring pairs $(1, 2), (3, 4), \ldots$. These are called odd layers. Alternating even and odd layers lets amplitude propagate across all modes. Schematically,

$$U \approx B_L^{\text{odd/even}} \cdots B_2^{\text{odd}} B_1^{\text{even}}, \tag{20}$$

with enough layers to realize an arbitrary dense $N \times N$ unitary.

Odd layers can be handled by shifting the target basis so that odd pairs become even pairs. If

$$S\, |y\rangle = |y + 1 \bmod N\rangle\,, \tag{21}$$

then an odd layer can be implemented as

$$B^{\text{odd}} = S^\dagger B^{\text{even}} S, \tag{22}$$

up to the chosen boundary convention.

1.9 Why two-mode operations can realize an arbitrary unitary

The most important missing fact is this: two-mode operations are not merely a convenient circuit family; they are *universal for finite-dimensional unitaries.* The reason is the same linear-algebra fact behind QR decomposition: any matrix can be triangularized by a sequence of plane rotations, also called Givens rotations.

Let

$$U = \sum_{r,c=0}^{N-1} U_{r,c} \, |r\rangle\langle c| \tag{23}$$

be an arbitrary $N \times N$ unitary. A two-mode operation acting on rows p and q from the left has the form

$$G_{p,q}^{\dagger} U, \tag{24}$$

where $G_{p,q}$ is identity except on the span of $|p\rangle, |q\rangle$. We choose $G_{p,q}$ so that the multiplication zeroes one selected entry of U while leaving all rows except p and q unchanged.

Concretely, suppose we want to zero the entry in row q of column c using rows p and q. Let

$$a = U_{p,c}, \qquad b = U_{q,c}, \qquad r = \sqrt{|a|^2 + |b|^2}. \tag{25}$$

If $r \neq 0$, define the 2×2 unitary

$$\widetilde{G}(a,b) = \frac{1}{r} \begin{bmatrix} a & b \\ -b^* & a^* \end{bmatrix}. \tag{26}$$

Then

$$\widetilde{G}(a,b) \begin{bmatrix} a^* \\ b^* \end{bmatrix} = \begin{bmatrix} r \\ 0 \end{bmatrix}, \tag{27}$$

or, with the conjugation convention adjusted appropriately, one obtains a two-row unitary that maps the pair of entries (a, b) to $(r, 0)$. The exact placement of conjugates depends on whether the operation is applied from the left or from the right, but the invariant statement is simple:

> Given two complex numbers a, b, there is a 2×2 unitary acting only on those two coordinates that maps (a, b) to $(\sqrt{|a|^2 + |b|^2}, 0)$.

Thus a two-mode operation can eliminate one matrix element without touching the other modes.

Now apply this repeatedly. Starting from the first column, use two-mode operations on rows $(N-2, N-1)$, then $(N-3, N-2)$, and so on, to zero all entries below the first row in column 0. Then move to column 1 and zero all entries below row 1, continuing until the matrix is upper triangular. Symbolically, there exists a product of two-mode unitaries $G_1, G_2, \ldots, G_K$ such that

$$G_K^{\dagger} \cdots G_2^{\dagger} G_1^{\dagger} U = R, \tag{28}$$

where R is upper triangular.

Because U is unitary and the G_i are unitary, R is also unitary. But a matrix that is both upper triangular and unitary must be diagonal. Therefore

$$R = D = \mathrm{diag}(e^{i\gamma_0}, e^{i\gamma_1}, \ldots, e^{i\gamma_{N-1}}), \tag{29}$$

and hence

$$U = G_1 G_2 \cdots G_K D. \tag{30}$$

This is the basic mathematical reason that a general unitary can be synthesized from two-mode operations plus a final diagonal phase layer.

1.10 A small 4×4 example

For $N = 4$, one possible elimination pattern is

$$G_{0,1}^{\dagger} G_{1,2}^{\dagger} G_{2,3}^{\dagger} U = \begin{bmatrix} * & * & * & * \\ 0 & * & * & * \\ 0 & * & * & * \\ 0 & * & * & * \end{bmatrix}, \tag{31}$$

where the three two-mode operations have removed the lower entries of column 0. Next use operations on rows $(2, 3)$ and $(1, 2)$ to remove the lower entries of column 1:

$$G_{1,2}^{\dagger} G_{2,3}^{\dagger} G_{0,1}^{\dagger} G_{1,2}^{\dagger} G_{2,3}^{\dagger} U = \begin{bmatrix} * & * & * & * \\ 0 & * & * & * \\ 0 & 0 & * & * \\ 0 & 0 & * & * \end{bmatrix}. \tag{32}$$

Finally remove the lower entry of column 2:

$$G_{2,3}^{\dagger} G_{1,2}^{\dagger} G_{2,3}^{\dagger} G_{0,1}^{\dagger} G_{1,2}^{\dagger} G_{2,3}^{\dagger} U = \begin{bmatrix} e^{i\gamma_0} & 0 & 0 & 0 \\ 0 & e^{i\gamma_1} & 0 & 0 \\ 0 & 0 & e^{i\gamma_2} & 0 \\ 0 & 0 & 0 & e^{i\gamma_3} \end{bmatrix}, \tag{33}$$

where the final equality uses unitarity: once all entries below the diagonal have been removed, the remaining upper-triangular unitary is diagonal. Reversing the order gives U as a product of nearest-neighbor two-mode operations and a final diagonal phase layer.

1.11 From arbitrary pairs to the rectangular nearest-neighbor mesh

The elimination argument above allows any pair of rows (p, q). The rectangular interferometer uses a more structured set of nearest-neighbor pairs:

$$(0, 1), (2, 3), \ldots \qquad \text{and} \qquad (1, 2), (3, 4), \ldots. \tag{34}$$

This still suffices because nearest-neighbor two-mode operations can move amplitude information across the whole line of modes. A non-neighbor operation on modes (p, q) can be compiled into nearest-neighbor operations by swapping or rotating amplitude along the path

$$p, p + 1, p + 2, \ldots, q. \tag{35}$$

The rectangular mesh is a scheduled version of this idea: it performs many disjoint nearest-neighbor eliminations in parallel, alternating even and odd pairings so that every neighboring edge is available at the correct stage of the elimination.

Thus the overall logic is:

$$\text{Givens/QR elimination} \Longrightarrow \text{arbitrary unitary} = \text{two-mode operations} + \text{diagonal phases}, \tag{36}$$

$$\text{nearest-neighbor scheduling} \Longrightarrow \text{two-mode operations can be arranged in even/odd layers}, \tag{37}$$

$$\text{phase-Hadamard decomposition} \Longrightarrow \text{each two-mode operation becomes phases plus fixed Hadamards}. \tag{38}$$

This is why the method is able to start from an arbitrary dense unitary and end with the layered beamsplitter circuit used later in the block-unitary construction.

1.12 Why the phase formulation is cheaper

The key cost-saving idea is now visible:

- The hard-looking dense unitary is decomposed into many simple two-mode operations.
- Each two-mode operation is expressed as phase shifts plus Hadamards.
- Hadamards do not contribute to the Toffoli count.
- Phase shifts can be batched: a single QROM load can output the phase data for many controlled phase rotations.
- Adjacent phase layers merge because diagonal matrices multiply by adding their phases.

Thus the repeated primitive is not "prepare a full column state and reflect about it," but rather "load a table of phases, apply cheap phase/Hadamard operations, and uncompute the table."

2 Problem statement

Let the selector register have dimension

$$M = 2^m,$$

and let the target register have dimension

$$N = 2^n.$$

We are given arbitrary dense target-register unitaries

$$U_x \in \mathbb{C}^{N\times N}, \qquad x \in \{0, 1, \ldots, M-1\}.$$

The desired controlled or block-diagonal unitary is

$$W = \sum_{x=0}^{M-1} |x\rangle\langle x| \otimes U_x. \tag{39}$$

Equivalently, on computational basis states,

$$W\,|x\rangle\,|y\rangle = |x\rangle\; U_x\,|y\rangle\,. \tag{40}$$

The selector x must remain unchanged. It is only a classical-looking quantum address that determines which target unitary is applied.

2.1 Why not synthesize W as a generic unitary?

A generic synthesis of W would treat it as a dense unitary of dimension MN. That ignores the fact that W has no couplings between different selector values:

$$\langle x'|\,\langle y'|\,W\,|x\rangle\,|y\rangle = 0 \qquad \text{whenever } x' \neq x.$$

The block-aware method below preserves this sparsity. It never applies operations that mix different x values. Instead, it applies a common target-register interferometer pattern whose angles are functions of x.

3 High-level structure

For each fixed x, decompose U_x using a rectangular interferometer synthesis:

$$U_x = D_x^{\text{fin}} \prod_{\ell=L}^{1} B_{x,\ell}, \tag{41}$$

where each $B_{x,\ell}$ is a layer of disjoint two-mode beam splitters acting on the target modes only, and D_x^{fin} is a final diagonal phase layer.

The block unitary is then

$$W = \sum_x |x\rangle\langle x| \otimes U_x \tag{42}$$

$$= \left(\sum_x |x\rangle\langle x| \otimes D_x^{\text{fin}}\right) \prod_{\ell=L}^{1} \left(\sum_x |x\rangle\langle x| \otimes B_{x,\ell}\right). \tag{43}$$

The important implementation principle is:

> Use the same layer topology for every x, but load different phase angles from QROM depending on x.

4 Step 1: choose target pairings

For N target modes, define even and odd pairings. The even pairing is

$$\mathcal{P}_{\text{even}} = \{(0,1),(2,3),(4,5),\ldots,(N-2,N-1)\}. \tag{44}$$

The odd pairing is

$$\mathcal{P}_{\text{odd}} = \{(1,2),(3,4),\ldots,(N-3,N-2)\}, \tag{45}$$

with boundary conventions handled by shifts or by omitting the unpaired endpoint.

A layer $B_{x,\ell}$ is block diagonal over one of these pairings. For an even layer,

$$B_{x,\ell}^{\text{even}} = \bigoplus_{j=0}^{N/2-1} G_{x,\ell,j}, \tag{46}$$

where $G_{x,\ell,j} \in \mathbb{C}^{2\times 2}$ acts on target modes $(2j, 2j+1)$. For an odd layer,

$$B_{x,\ell}^{\text{odd}} = \bigoplus_{j} G_{x,\ell,j} \quad \text{on target pairs } (2j+1, 2j+2). \tag{47}$$

The same sequence of even and odd layer types is used for every x.

5 Step 2: classically compute the beam-splitter parameters

For every selector value x, run the classical rectangular-interferometer decomposition on U_x. This produces parameters for each two-mode block

$$G_{x,\ell,j}.$$

A generic two-mode unitary can be written, up to convention-dependent global and one-sided phases, as phases plus Hadamards:

$$G_{x,\ell,j} = D^{(a)}(\alpha_{x,\ell,j}) H D^{(b)}(\beta_{x,\ell,j}) H D^{(c)}(\chi_{x,\ell,j}), \tag{48}$$

where

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \tag{49}$$

and a one-sided phase convention may be chosen as

$$D(\theta) = \begin{bmatrix} e^{i\theta} & 0 \\ 0 & 1 \end{bmatrix}. \tag{50}$$

The exact assignment of α, β, χ depends on the chosen beam-splitter convention. What matters for synthesis is that the non-Clifford work is converted into diagonal phase rotations, while the Hadamards are Clifford gates.

6 Step 3: merge adjacent phase layers

Adjacent diagonal phase layers multiply entrywise. If

$$D_a = \sum_y e^{ia_y} |y\rangle\langle y| , \qquad D_b = \sum_y e^{ib_y} |y\rangle\langle y| , \tag{51}$$

then

$$D_a D_b = \sum_y e^{i(a_y+b_y)} |y\rangle\langle y| . \tag{52}$$

Thus, neighboring phase layers from adjacent beam-splitter decompositions can be absorbed into one another.

For block unitaries, the same identity holds pointwise in x:

$$\left(\sum_{x,y} e^{ia_{x,y}} |x\rangle\langle x| \otimes |y\rangle\langle y|\right)\left(\sum_{x,y} e^{ib_{x,y}} |x\rangle\langle x| \otimes |y\rangle\langle y|\right) \tag{53}$$

$$= \sum_{x,y} e^{i(a_{x,y}+b_{x,y})} |x\rangle\langle x| \otimes |y\rangle\langle y| . \tag{54}$$

After merging, each interferometer layer can be implemented using a small number of multiplexed phase functions such as

$$\alpha_\ell(x,j), \qquad \beta_\ell(x,j), \tag{55}$$

plus a final phase function

$$\gamma(x,y). \tag{56}$$

7 Step 4: implement an even layer

For an even layer, write the target index as

$$y = 2j + s, \qquad s \in \{0,1\}, \qquad j \in \{0,\dots,N/2-1\}. \tag{57}$$

That is, the target register is decomposed as

$$|y\rangle = |j\rangle\,|s\rangle , \tag{58}$$

where s is the least significant target bit and j contains the remaining $n-1$ target bits.

A block-controlled even layer has the form

$$\mathcal{M}_\ell^{\text{even}} = \sum_{x=0}^{M-1}\sum_{j=0}^{N/2-1} |x\rangle\langle x| \otimes |j\rangle\langle j| \otimes V_\ell(x,j), \tag{59}$$

where the active one-qubit operation on s is

$$V_\ell(x,j) = R(\alpha_\ell(x,j))\, H\, R(\beta_\ell(x,j))\, H. \tag{60}$$

Here

$$R(\theta) = \begin{bmatrix} e^{i\theta} & 0 \\ 0 & 1 \end{bmatrix} \tag{61}$$

is a phase rotation in a one-sided convention. Other equivalent Z-rotation conventions may be used, with corresponding changes to the stored angles.

Therefore the action on basis states is

$$|x\rangle\,|j\rangle\,|s\rangle \longmapsto |x\rangle\,|j\rangle\; V_\ell(x,j)\,|s\rangle . \tag{62}$$

8 Step 5: QROM-load the layer phases

For each layer ℓ, create a QROM table indexed by (x, j):

$$(x, j) \longmapsto (\alpha_\ell(x, j), \beta_\ell(x, j)) \,. \tag{63}$$

The coherent QROM load is

$$\mathrm{QROM}_\ell : |x\rangle\,|j\rangle\,|0\rangle\,|0\rangle \longmapsto |x\rangle\,|j\rangle\,|\alpha_\ell(x, j)\rangle\,|\beta_\ell(x, j)\rangle \,. \tag{64}$$

The angle registers hold b-bit approximations. Once the angles are loaded, apply phase-gradient rotations to the active qubit:

$$|\alpha\rangle\,|\beta\rangle\,|s\rangle \longmapsto |\alpha\rangle\,|\beta\rangle\; R(\alpha)HR(\beta)H\,|s\rangle \,. \tag{65}$$

Then erase the QROM output:

$$\mathrm{QROM}_\ell^\dagger : |x\rangle\,|j\rangle\,|\alpha_\ell(x, j)\rangle\,|\beta_\ell(x, j)\rangle \longmapsto |x\rangle\,|j\rangle\,|0\rangle\,|0\rangle \,. \tag{66}$$

Thus one even layer is implemented as

$$\mathcal{M}_\ell^{\mathrm{even}} = \mathrm{QROM}_\ell^\dagger\,[R(\alpha)HR(\beta)H]\,\mathrm{QROM}_\ell, \tag{67}$$

where the bracketed operation uses the loaded angle registers.

9 Step 6: implement odd layers using target-only shifts

Odd layers couple target modes $(1, 2), (3, 4), \ldots$. Rather than use a different circuit topology, convert an odd pairing into an even pairing by shifting the target register.

Define the target-register shift

$$S\,|y\rangle = |y + 1 \bmod N\rangle \,. \tag{68}$$

Then an odd layer can be implemented as a conjugated even layer:

$$\mathcal{M}_\ell^{\mathrm{odd}} = (\mathbb{I}_x \otimes S^\dagger)\mathcal{M}_\ell^{\mathrm{even}}(\mathbb{I}_x \otimes S), \tag{69}$$

or with S and $S^\dagger$ interchanged depending on the indexing convention.

The crucial constraint is that S acts only on the target register:

$$|x\rangle\,|y\rangle \xmapsto{\mathbb{I}_x \otimes S} |x\rangle\,|y + 1 \bmod N\rangle \,. \tag{70}$$

It must not act as

$$|x, y\rangle \mapsto |x, y + 1 \bmod MN\rangle \tag{71}$$

under an encoding of the combined register. Such a combined shift would incorrectly mix different x-blocks.

10 Step 7: apply the final block-dependent diagonal layer

After the beam-splitter layers, each block has a final diagonal phase. For the block unitary this is

$$D_{\text{fin}} = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} e^{i\gamma(x,y)} \left|x\right\rangle\!\left\langle x\right| \otimes \left|y\right\rangle\!\left\langle y\right| . \tag{72}$$

This is implemented by one more QROM table,

$$(x, y) \longmapsto \gamma(x, y), \tag{73}$$

with coherent load

$$\left|x\right\rangle \left|y\right\rangle \left|0\right\rangle \longmapsto \left|x\right\rangle \left|y\right\rangle \left|\gamma(x, y)\right\rangle . \tag{74}$$

Apply the phase

$$\left|\gamma(x, y)\right\rangle \left|x\right\rangle \left|y\right\rangle \longmapsto e^{i\gamma(x,y)} \left|\gamma(x, y)\right\rangle \left|x\right\rangle \left|y\right\rangle , \tag{75}$$

and then uncompute the angle register.

A subtle point is that x-dependent global phases of the blocks are not globally irrelevant for the direct sum. If

$$U_x \mapsto e^{i\delta_x} U_x, \tag{76}$$

then

$$W \mapsto \sum_x e^{i\delta_x} \left|x\right\rangle\!\left\langle x\right| \otimes U_x, \tag{77}$$

which changes relative phases between selector branches. Therefore the final phase layer must retain block-dependent phase information.

11 Step 8: assemble the full circuit

Let S_ℓ denote either $\mathbb{I}$ or S depending on whether layer ℓ is already even or must be shifted into the even pairing. Then the block-unitary circuit can be written as

$$W = D_{\text{fin}} \prod_{\ell=L}^{1} \left[(\mathbb{I}_x \otimes S_\ell^\dagger) \mathcal{M}_\ell (\mathbb{I}_x \otimes S_\ell) \right] , \tag{78}$$

where

$$\mathcal{M}_\ell = \sum_{x,j} \left|x\right\rangle\!\left\langle x\right| \otimes \left|j\right\rangle\!\left\langle j\right| \otimes \left[R(\alpha_\ell(x, j)) H R(\beta_\ell(x, j)) H \right] . \tag{79}$$

Equivalently, using QROM explicitly,

$$\mathcal{M}_\ell = \text{QROM}_\ell^\dagger \left[R(\alpha) H R(\beta) H \right] \text{QROM}_\ell, \tag{80}$$

where

$$\text{QROM}_\ell \left|x\right\rangle \left|j\right\rangle \left|0\right\rangle \left|0\right\rangle = \left|x\right\rangle \left|j\right\rangle \left|\alpha_\ell(x, j)\right\rangle \left|\beta_\ell(x, j)\right\rangle . \tag{81}$$

This construction gives

$$\left|x\right\rangle \left|y\right\rangle \longmapsto \left|x\right\rangle U_x \left|y\right\rangle , \tag{82}$$

as required.

12 Step 9: Toffoli cost model

Let b be the number of bits used to store each phase angle. Let Λ be the usual QROM tradeoff parameter. For a non-block dense $N \times N$ unitary, the phasing-layer part of the interferometer synthesis has the form

$$T_{\text{phase}}(N) \approx N\left(\left\lceil \frac{N}{2\Lambda} \right\rceil + 2\Lambda b - 5\right), \tag{83}$$

up to lower-order shift and final-phase terms.

For the block unitary, the QROM address is enlarged from j to (x, j). Each layer table has approximately $MN/2$ entries instead of $N/2$ entries. Therefore the analogous leading cost is

$$T_{\text{phase}}^{\text{block}}(M, N) \approx N\left(\left\lceil \frac{MN}{2\Lambda} \right\rceil + 2\Lambda b - 5\right). \tag{84}$$

The final phase layer has approximately MN entries, with cost of the form

$$T_{\text{fin}}^{\text{block}} \approx \left\lceil \frac{MN}{\Lambda_f} \right\rceil + \Lambda_f b + \left\lceil \frac{MN}{\Lambda'_f} \right\rceil + \Lambda'_f - 6. \tag{85}$$

The target-register shifts are not multiplied by M, because they are shared across selector branches:

$$T_{\text{shift}} \approx (n-2)(N-1). \tag{86}$$

A practical block-aware estimate is therefore

$$T_{\text{block}} \approx N\left(\left\lceil \frac{MN}{2\Lambda} \right\rceil + 2\Lambda b - 5\right) + (n-2)(N-1) + T_{\text{fin}}^{\text{block}}. \tag{87}$$

12.1 Optimizing the QROM parameter

Ignoring ceilings and constants, the leading term is

$$N\left(\frac{MN}{2\Lambda} + 2\Lambda b\right). \tag{88}$$

Minimizing over Λ gives

$$\frac{d}{d\Lambda}\left(\frac{MN}{2\Lambda} + 2\Lambda b\right) = 0, \tag{89}$$

so

$$-\frac{MN}{2\Lambda^2} + 2b = 0, \tag{90}$$

and hence

$$\Lambda_{\text{opt}} \approx \frac{1}{2}\sqrt{\frac{MN}{b}}. \tag{91}$$

Substituting this scaling into the leading term gives

$$T_{\text{phase}}^{\text{block}} = O\left(N\sqrt{MNb}\right) = O\left(\sqrt{M}\, N^{3/2}\sqrt{b}\right). \tag{92}$$

13 Step 10: compare three strategies

There are three natural approaches.

13.1 Strategy A: synthesize each block independently

If each U_x is synthesized separately, the cost is roughly

$$T_{\text{separate}} \approx M\, T_{\text{dense}}(N) = O\Big(MN^{3/2}\sqrt{b}\Big)\,. \tag{93}$$

13.2 Strategy B: synthesize W as a generic dense unitary

If W is treated as a generic dense unitary of dimension MN, the cost is roughly

$$T_{\text{generic}} = O\Big((MN)^{3/2}\sqrt{b}\Big) = O\Big(M^{3/2}N^{3/2}\sqrt{b}\Big)\,. \tag{94}$$

This is wasteful because it ignores the absence of couplings between different x sectors.

13.3 Strategy C: block-aware multiplexed interferometer

The method in this note gives

$$T_{\text{block}} = O\Big(\sqrt{M}\, N^{3/2}\sqrt{b}\Big)\,. \tag{95}$$

Thus, at the scaling level,

$$\frac{T_{\text{separate}}}{T_{\text{block}}} = O(\sqrt{M}), \tag{96}$$

and

$$\frac{T_{\text{generic}}}{T_{\text{block}}} = O(M). \tag{97}$$

The savings arise because the interferometer skeleton is shared across all blocks, while only the QROM-loaded angles depend on x.

14 Algorithm summary

Step 1: Input data. Receive M dense $N \times N$ unitaries U_x.

Step 2: Classical decomposition. For each x, compute a rectangular-interferometer decomposition

$$U_x = D_x^{\text{fin}} \prod_{\ell=L}^{1} B_{x,\ell}.$$

Step 3: Use common topology. Ensure that every $B_{x,\ell}$ uses the same even/odd layer schedule. Only the angles vary with x.

Step 4: Convert two-mode blocks. Rewrite each $G_{x,\ell,j}$ as phases and Hadamards:

$$G_{x,\ell,j} = D^{(a)} H D^{(b)} H D^{(c)}.$$

Step 5: Merge diagonal phases. Combine adjacent diagonal layers by adding their phases:

$$e^{ia_{x,y}} e^{ib_{x,y}} = e^{i(a_{x,y}+b_{x,y})}.$$

Step 6: Build QROM tables. For each layer, store

$$(x, j) \mapsto (\alpha_\ell(x, j), \beta_\ell(x, j)).$$

For the final diagonal layer, store

$$(x, y) \mapsto \gamma(x, y).$$

Step 7: Implement even layers. For $y = 2j + s$, use QROM to load $\alpha_\ell(x, j), \beta_\ell(x, j)$ and apply

$$R(\alpha_\ell(x, j))HR(\beta_\ell(x, j))H$$

to the active bit s.

Step 8: Implement odd layers. Conjugate by target-only shifts:

$$\mathcal{M}_\ell^{\text{odd}} = (\mathbb{I}_x \otimes S^\dagger)\mathcal{M}_\ell^{\text{even}}(\mathbb{I}_x \otimes S).$$

Step 9: Apply final phases. Use QROM addressed by (x, y) to apply

$$e^{i\gamma(x,y)}.$$

Step 10: Output. The resulting unitary is

$$|x\rangle |y\rangle \mapsto |x\rangle U_x |y\rangle .$$

15 Practical caveats

Independent dense blocks still require large classical data. The method reduces Toffoli cost by sharing a circuit skeleton, but it does not reduce the information content of arbitrary unrelated dense unitaries. In the worst case, the classical angle table size is

$$O(MN^2). \tag{98}$$

Block-dependent phases matter. A phase that is global inside one block is not necessarily global for the whole direct sum. Phases depending on x must be retained unless one knows that the selector register is always classical and never appears in coherent superposition.

The selector register is an address, not a mode. All beam splitters, Hadamards, and shifts act on the target register. The selector register controls only which phase angles are loaded.

The method is most useful for dense target blocks. If the U_x have special structure, such as being sparse, diagonal, tensor-product, Clifford, low-rank, or generated by short Hamiltonian evolution, a specialized synthesis may beat dense interferometer synthesis.

16 References

- Interferometer-based dense unitary synthesis paper: https://arxiv.org/pdf/2409.11748
- Low–Kliuchnikov–Schaeffer (LKS) state-preparation/unitary-synthesis paper: https://arxiv.org/pdf/1812.00954